

CHANITNAN KITNANTAKHUN

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SUMMARY

Highly motivated Backend Developer with specialized expertise in system architecture and robust API design. Proficient in Spring Boot, C and Python, ensuring data durability and fault tolerance through custom-built financial and caching systems. Specializes in low-level concurrency, secure data handling, and collaborating with cross-functional teams to deliver impactful software products.

TECHNICAL SKILLS

- **Frontend:** HTML · CSS (Bootstrap 5 & Tailwind CSS) · JavaScript · TypeScript · React
- **Backend & Database:** Python (Django & FastAPI) · Java (Spring Boot) · JWT · PostgreSQL · Redis · SQLite
- **DevOps & Cloud:** Docker · AWS S3 · CloudFront · EC2 · Railway · Nginx · Apache2 · Flyway · GitHub Actions
- **Tools & Environment:** Git · Zsh · Bruno · Vite · npm · Bun · Make · Figma

PROJECTS

AgriscanPro | Django · PostgreSQL · Redis · React · TypeScript · Tailwind CSS · AWS CloudFront · S3 · Railway · GitHub Actions

- Designed comprehensive RESTful API covering sample tracking, process logs, and mycotoxin results; implemented JWT authentication for stateless API security
- Engineered Role-Based Access Control (RBAC) with 6 distinct roles via custom User Models and permission classes
- Deployed backend on Railway (PostgreSQL + Django) and frontend on AWS S3 + CloudFront CDN; automated deployment pipeline with GitHub Actions CI/CD
- Implemented in-browser CSV and Excel (XLSX) export with server-side pagination and complex filtering on dynamic data tables using TanStack Query and Tailwind CSS
- End-to-end web application for agricultural sample management built with food science researchers at university responsible for system architecture design, API backend, and React SPA frontend

TPSystem | Spring Boot · Java 17 · PostgreSQL · Flyway · JWT · Redis · Docker · React · TypeScript · Nginx

- Implemented ACID transactions with Pessimistic locking (SELECT FOR UPDATE) and lock ordering to prevent deadlocks; built Idempotency key pattern to eliminate duplicate charges on network retry
- Engineered JWT auth with refresh token rotation, TOTP-based 2FA (RFC 6238), and sliding window rate limiting — preventing replay attacks and credential stuffing
- Containerized full stack with Docker Compose (multi-stage builds); Flyway for versioned DB migrations; Nginx as reverse proxy
- Designed append-only Audit Trail with hash chaining for tamper detection; implemented read-your-writes consistency to prevent stale balance reads
- Banking transaction processing system built phase-by-phase to reproduce real failure modes (race conditions, duplicate payments, cascading failures) then implement the patterns banks actually use to solve them

Mini Redis | C (POSIX) · TCP/IP · RESP Protocol · Make · Linux

- Engineered custom Hash Map using a fast string hashing algorithm with Dynamic Resizing — O(1) average time complexity for SET, GET, DEL
- Implemented I/O Multiplexing with poll() to handle multiple concurrent clients in a single thread — zero busy-wait overhead
- Designed Append-Only File (AOF) persistence using fsync syscall — every write logged to disk for full crash recovery on restart
- In-memory key-value store written in C from scratch, inspired by Redis — built after studying TCP/IP internals through Beej's Guide to Network Programming

EDUCATION

Thammasat University 2025 - Present
B.Sc. Computer Science 3.5 GPAX

CERTIFICATION & AWARDS

- **AWS Academy Cloud Developing (In Progress) : S3 · DynamoDB · Lambda · API Gateway · SQS/SNS · CloudFront** 2026
- **Excellent in Teamwork Award - PRAGMA 41 x SEAIP Hackathon** 2026
International Hackathon · AI for Health Innovation · Thammasat University x Yarsi University
Project: AI-based Decision-Support System for Sepsis Management in ICU
- **DataCamp Courses:** Supervised Learning with scikit-learn · Introduction to FastAPI Introduction to NumPy 2025
Data Manipulation with pandas · Intermediate Git

ADDITIONAL INFO

- **Undergraduate Researcher** : Thammasat University Tonkit Lab 2025 - Present
- **Languages:** Thai (Native) · English (Intermediate)
- **Interests:** Systems Programming · Distributed Systems · Open Source